NagarSetu: A Bilingual AI-Powered System for Smart Civic Complaint Management and RTI Assistance

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Abstract*—The growth in volume of civic complaints is a result of rapid urbanization, while existing grievance redressal mechanisms still work manually, language-limited, and inefficient. At the same time, the Right to Information (RTI) Act, though essential for transparency, is not effectively used because writing formal requests is often complex. This paper presents NagarSetu, a bilingual, multimodal, AI-powered civic engagement platform designed to automate complaint handling and assist citizens in RTI drafting. The system supports text, image, and voice inputs in English and Hindi. Image-based complaints are classified using a lightweight MobileNet-based convolutional neural network, while text and translated voice complaints are processed using a BERT/DistilBERT-based classifier. Whisper speech recognition framework is used to convert speech inputs to text, and RTI drafts are generated using a Retrieval-Augmented Generation (RAG) pipeline grounded in RTI Act provisions. Experimental evaluation shows that the proposed model achieves an overall accuracy of approximately 89%, while significantly reducing routing latency and improving user satisfaction compared to manual and keyword-based systems.

Keywords*—Civic Grievance Redressal, Smart City Governance, Artificial Intelligence, Natural Language Processing, Computer Vision, Speech Recognition, RTI Automation, Bilingual Systems Introduction

I. INTRODUCTION

Sustainable and livable cities that directly affect the quality of life and daily experiences of citizens are shaped by urban governance. Rapid urbanization and population growth have placed significant pressure on urban infrastructure [1], [2], public services, and administrative systems. It is mandatory for the Municipal authorities to address diverse civic issues including poor road conditions, waste management inefficiencies, inconsistent water supply, and public safety concerns. The increasing scale and complexity of these challenges demand governance systems that are responsive, adaptive, and citizen-centric.

An effective grievance redressal mechanism acts as a vital communication bridge between citizens and municipal authorities. It allows residents to report civic issues, request timely intervention, and ensure accountability in service delivery. Such efficient grievance systems contribute to improved public services, enhanced citizen trust, and informed urban planning. However, in many cities, grievance handling remains slow, disconnected, and inefficient, which results in inadequacy of such platforms [3].

A. Background: Urban Governance, Smart Cities, Grievance Redressal, and the RTI Act

Urban governance focuses on the planning, coordination, and management of civic resources to ensure fair development and improved quality of life. Transparency, accountability, and citizen participation are critical elements of effective governance. Smart city initiatives increasingly influence digital technologies such as online portals, mobile applications, and data-driven systems to enhance service delivery and communication between citizens and authorities [1], [5].

Digital grievance redressal platforms allow citizens to report issues favorably without visiting municipal offices. The goal for these systems is to reduce administrative overhead, improve response times, and enable structured complaint tracking. When effectively implemented, they enhance transparency by keeping citizens informed about grievance status and resolution actions [2]. The Right to Information (RTI) Act further strengthens democratic accountability by enabling citizens to request information from government departments. Informed civic participation that allows citizens to monitor administrative decisions and public spending is supported by RTI mechanisms [7]. However, limited awareness, complex procedures, and accessibility challenges restrict the effective use of both grievance portals and RTI systems, reducing their overall impact [4].

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B. *Practical Challenges: Manual Systems, Delays, and Lack of Transparency*

Many systems still continue to rely on manual processes for complaint review, classification, and routing. This dependence leads to delays, inconsistent handling, and prolonged resolution timelines, despite the availability of online grievance portals. Paper-based documentation, delayed inspections, and weak coordination between departments further reduce system efficiency and citizen trust [3], [5]. Most platforms provide limited or delayed status updates; that leads to another major concern, transparency. Language barriers restrict participation, with most systems operating mainly in English, excluding non-English-speaking citizens. Additionally, inadequate support for image and voice-based inputs limits accessibility for elderly and non-technical users, reducing the clarity and authenticity of reported issues [2], [4].

C. *Motivation and Significance of Automating Complaint and RTI Workflows*

The automation of grievance redressal and RTI workflows is driven by the need to minimize human dependency, improve response times, and enhance accessibility. Automated systems can accurately classify and route complaints, ensuring timely delivery to appropriate departments. Intelligent prioritization enables authorities to address critical and high-risk issues more effectively [1], [6]. Automating RTI processes is equally significant, as many citizens are discouraged by the complexity of drafting formal RTI applications. Guided and automated RTI drafting assistance can simplify this process, encourage civic engagement, and strengthen transparency. Together, automation and intelligent assistance can transform grievance redressal into a more proactive and citizen-centric governance process [7].

D. *Contributions of NagarSetu*

To overcome the identified challenges, NagarSetu is proposed as a bilingual and multimodal AI-driven civic assistance platform. The system enables citizens to submit grievances using text, images, or voice inputs in both English and Hindi. It integrates natural language processing for text analysis, computer vision for image-based issue identification, and speech recognition for voice input processing. Additionally, NagarSetu incorporates retrieval-augmented generation to assist users in drafting structured RTI applications. The key contribution of NagarSetu lies in its unified integration of multiple AI technologies to support automated complaint handling, RTI assistance, real-time tracking, and enhanced transparency. By simplifying interactions between citizens and municipal authorities, the platform aims to improve governance efficiency, increase citizen engagement, and contribute to the broader goals of smart and inclusive urban development.

II. LITERATURE REVIEW

A. *Existing complaint management and e-governance systems and their limitations*

Existing digital grievance redressal and e-governance systems have been developed to improve accessibility and reduce the dependence on physical complaint submission. These systems commonly include web portals and mobile applications that allow citizens to register complaints related to urban infrastructure and public services. While such platforms have improved the ease of reporting, most continue to rely on manual workflows for complaint verification, categorization, and departmental routing. This results in delays, inconsistent handling of grievances, and limited scalability when complaint volumes increase [1], [2].

Many smart city portals incorporate basic features such as geolocation tagging and complaint tracking; however, they often lack intelligent automation. Complaint classification is typically performed by municipal staff, which increases response time and introduces the risk of misclassification. Additionally, most platforms restrict users to text-based inputs, limiting the ability to provide visual or audio evidence. Language support is also limited, as many systems operate only in English, reducing accessibility for non-English-speaking users and affecting inclusive participation [3], [4].

B. *AI in civic tech*

Recent advancements in artificial intelligence have enabled more efficient handling of civic complaints through automated analysis and decision support. Natural language processing techniques have been applied to classify textual complaints, using both traditional machine learning models and transformer-based architectures for improved accuracy [1], [2]. These approaches help reduce manual effort and improve routing efficiency but are often limited to text data.

Computer vision techniques, particularly convolutional neural networks, have been explored for image-based detection of civic issues such as potholes, garbage accumulation, and damaged infrastructure [5], [6]. Voice input systems using speech-to-text models have also been proposed to enhance accessibility for elderly and non-technical users. Despite these advancements, most AI-based solutions address individual modalities in isolation and are not integrated into end-to-end grievance management systems. The application of AI for RTI assistance remains limited, with existing systems providing minimal automation or contextual guidance.

C. *Identified research gap*

The review of existing literature highlights a clear gap in the development of a comprehensive civic engagement platform. Current solutions focus on isolated functionalities such as text classification, image detection, or basic complaint submission,

without unifying these capabilities into a single system. Furthermore, limited language support and the absence of automated RTI drafting tools restrict broader citizen participation.

There is a need for a unified, bilingual, and multimodal platform that integrates artificial intelligence across complaint reporting, classification, prioritization, and RTI assistance. Such a system should support text, image, and voice inputs while providing intelligent automation and transparency throughout the grievance redressal process. Addressing this gap forms the foundation of the proposed NagarSetu system.

Table I. Comparative Analysis of Existing Systems

System Type	Automation Level			Multimodal Support	Language Support		RTI Assistance
	Low	Medium	High		English	Hindi	
Traditional grievance portals	✔	✘	✘	No	✔	✘	No
Smart city mobile apps	✘	✔	✘	Limited (Text + Geo)	✔	✘	No
AI-based complaint classifiers	✘	✔	✔	Single modality	✔	✘	No
NagarSetu (Proposed)	✘	✘	✔	Text, Image, Voice	✔	✔	Yes

Table II. Literature Survey of the previous studies

Reference & Year	Domain	Algorithm	Dataset	Research Gap
[1] 2025	AI-Powered Petition Analysis & Grievance Management System	FastAPI backend, dashboards, ML for petitions	Citizen petition corpus	Limited RTI automation
[4] 2025	Digital Grievance Redressal for the Cleaner and Smarter India	Location tagging + AI classification	Case study dataset	No bilingual/ voice support
[3] 2025	CivicFix: Smart Complaint Routing for Urban Solutions	ML based complaint routing, NLP classification, Flow Management	City grievance datasets	Limited RTI integration, scalability constraints
[7] 2024	CitySolution: Mobile DL Complaint System	CNN-based image classification & monitoring	Android app complaint images	No RTI or Voice input
[8] 2021	Automated Complaint Classification & Routing Using NLP	TF-IDF, Logistic Regression & BERT	Public complaint datasets	No Image Classification
[12] 2021	AI-Driven Complaint Management System	Machine learning–based complaint classification, automated routing, decision support system	Citizen grievance data, Public grievance datasets	Limited explainability, restricted real-time deployment

III. METHODOLOGY

This section describes the architectural framework of NagarSetu, which shows how different components work together to support smart civic complaint handling and RTI assistance. The design mainly focuses on scalability, automation, and inclusiveness by combining artificial intelligence, geospatial services and role-based dashboards [1], [4], [8]. NagarSetu follows a multi-layered architecture with four connected layers: the Presentation Layer which provides interfaces for citizens, municipal staff and administrators [2], [7], [8]; the Intelligence Layer that handles AI-based classification, prioritization and RTI generation; the Integration Layer which connects external services like mapping and notification systems [4], [7]; and the Data Layer that stores complaints, RTI records and user activity in a centralized manner.

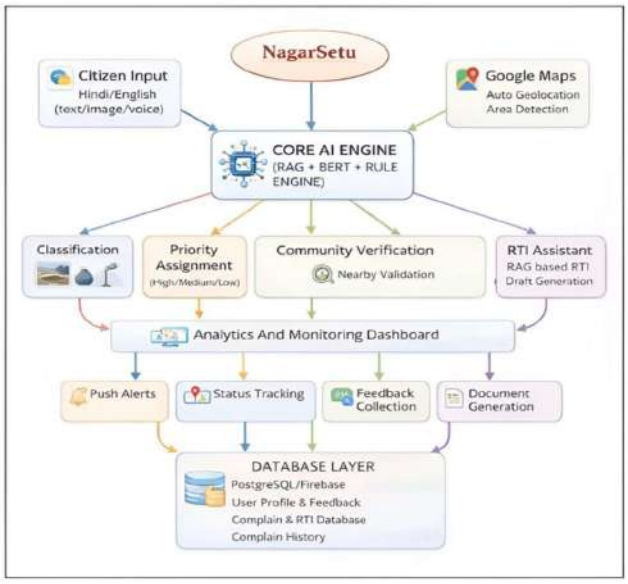

Fig. 1. System Architecture Diagram for NagarSetu

The proposed NagarSetu platform which explains in figure 1, emphasizing the interaction among system outputs, core processing modules, and citizen inputs. Centralized data management and AI-assisted mechanisms used to make easier gathering, categorization, prioritization, and solutions of civic complaints.

1) Citizen Input Layer

Interface accepts multi-modal inputs in the form of text, images, and voice recordings. It contains complaints related to public services and infrastructures. Key requirements in such platforms are satisfied in this system due to diverse input formats. This leads to improvement in accessibility and inclusivity [2], [4], [7], [8].

2) Core AI-Enabled Processing Layer

The center of architecture is Core AI Engine which includes Natural Language Processing, Computer Vision, and Rule-Based decisions. Functions performed by this layer are as follows:

1. Issue Classification: Complaints are categorized into predefined classes such as potholes, garbage, broken streetlights, water blockage. NLP and image classification models like MobileNet are used for detecting issues [9]– [11].
2. Priority Assignment: Based on severity, urgency, and public impact of each complaint priority level (High, Medium, or Low) is assigned. This helps admin to allocate workers in different locations.
3. Geo-Mapping and Area Detection: From images location metadata extracted, and then processed using map services. It helps in identifying exact geographical areas for location specific resolution of complaints.
4. Community Verification: Credibility and confidence scores increased due to neighboring community's validation. and for that nearby users verify complaints by giving reports.
5. RTI Assistant Module: System offers an RTI assistant for generating structured RTI applications draft using legal templates. This module supports compliance with the Right to Information Act, 2005 [5], [15].

3) Analytics and Monitoring Layer

Real-time insights to admins provided by monitoring dashboards with the help of processed data. status tracking used to monitor the lifecycle of complaints from submission to resolution, feedback and push alerts to users and officials regarding updates. These monitoring mechanisms are essential for improving accountability and service quality of grievance redressal systems [1], [4], [8].

4) Database Layer

User authentication records, complaint metadata, RTI drafts and history this system data is stored in a structured database layer. Centralized data storage helps in integrity, traceability, and long-term availability for auditing and reporting purposes [2], [7], [12].

5) System Outputs

For citizens it gives transparent status updates and generates RTI applications drafts. For admins it gives real-time alerts, notifications, and analytics as the primary outputs of the system. It establishes closed-loop feedback which improves transparency and public trust [3], [4], [8].

A. High-Level System Interaction Model

NagarSetu operates as a centralized digital grievance redressal platform that enables structured interaction among citizens, municipal authorities, and backend processing units. The system functions as an intermediary that receives complaints, processes them using predefined workflows, and communicates updates in real time, following interaction models proposed in prior smart governance frameworks [1], [2], [8]. Each complaint submission is registered with associated metadata such as location, issue category, and timestamp. Expected response timelines are computed based

on priority levels and predefined service benchmarks. Complaint records are synchronized with databases, and real-time updates are shared to backend services used for analytics, notifications, and monitoring [4], [7], [12].

B. Role-Based Operational Workflow

There are three primary stakeholders in the workflow of the system which follows role-based access.

User Operations
Users access the system through a secure login mechanism. After authentication users are allowed to register complaints by providing details. And users can monitor the progress of complaints after submission. It improves user trust by handling and resolution of reported complaints [4], [8].

Administrative Operations
Administrators supervised the system, after authentication they will review incoming complaints to verify validity and completeness. After verification, these complaints are assigned to specific workers according to issue type, location, and availability. It ensures proper allocation and prevents misuse of complaints.

Worker Operations
On interface workers receive assigned tasks and are responsible for on-ground execution and resolution of issues. Workers may submit updates or reports recorded in the system and reflected in the complaint status visible to both administrators and users [7], [8]. The process done by correct action and status updates and logging all details from submission and verification [1], [3], [8].

C. System Relevance

The system combines complaint submission, access controls, real time notifications, and data integration for transparent grievance redressal. Multimodal AI pipeline converts Hindi/English text, images, voice inputs into complaints, RTI drafts, tasks that are accessible through real-time dashboards enable data driven governance.
The end-to-end workflow can be written as:
Input acquisition: Users submit complaints in the form of text, image, voice through interface.
AI processing:

1. MobileNet-based classifier is used to identify issue types in images like potholes, garbage dumping, etc.
2. BERT classifier used to categorized text and transcribed voice complaints into department-specific categories and priority levels.
3. To enable low-literacy and useful for older users, voice inputs are converted to text using Whisper speech-to-text to interact with the system.
4. Retrieval-Augmented Generation (RAG) module used to handle RTI-related queries. That retrieves RTI Act sections and templates and generates draft applications.

Routing and storage: After classification and prioritization complaints are stored in Firebase Firestore along with geolocation data, and routed to a particular department.
Dashboards and feedback: Complaint status, maps, analytics and RTI drafts with notification is displayed on role-specific dashboards [8][9].

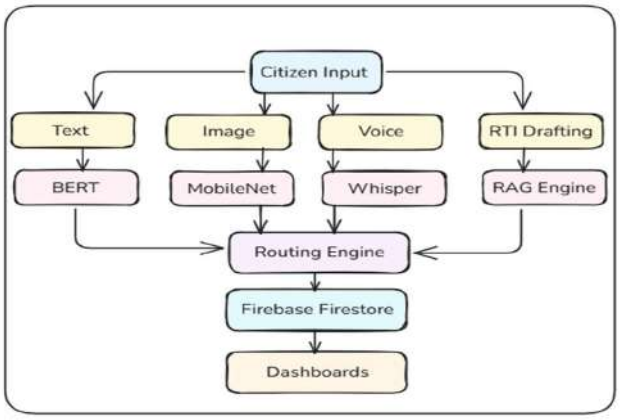

Fig. 2. High-level pipeline diagram

D. Datasets and preprocessing for images and text

1. Image dataset
   The image dataset for NagarSetu consists of civic-issue photographs drawn from publicly available pothole and garbage datasets and from field data collected using smartphone cameras in local neighbourhoods. Images cover common municipal problems including potholes, broken streetlights, garbage dumping, water leakage, and other visible infrastructure defects. Each image is assigned a single dominant label corresponding to the municipal department responsible for resolution (e.g., Roads, Solid Waste, Water Supply, Street-Lighting), consistent with prior civic complaint classification studies and pothole detection datasets [10]. Preprocessing resizes all images to 224×224 pixels to match MobileNet input requirements, normalizes pixel values using ImageNet mean and variance for stable optimization, and applies data augmentation including random horizontal flips, small rotations, and brightness variations to address differences in camera angle, lighting, and device quality [13].

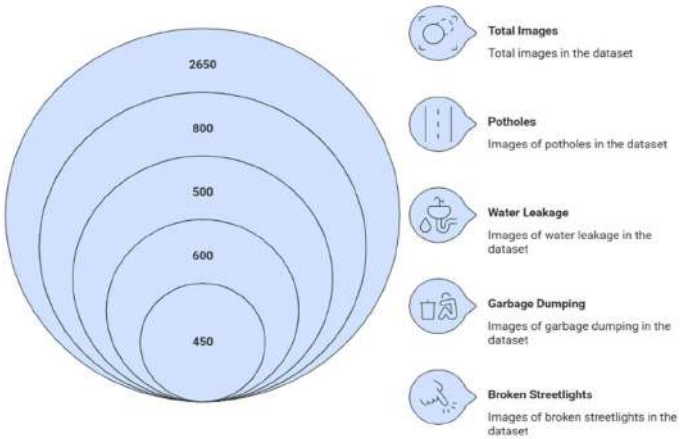

Fig. 3. Image Data Summary

2. Text and voice complaint dataset
The text dataset is composed of manually authored complaint descriptions, historical examples, and synthetically generated samples to cover under-represented categories such as water leakage or dangerous electrical faults. Complaints are written in English, Hindi, and Hindi–English code-mixed style to match real citizen language patterns. Voice complaints collected through the web interface are recorded as short audio clips, which are transcribed to text using the Whisper model configured for Indian English and Hindi. These transcripts are merged with text complaints to form a unified corpus for text classification. Text and transcript preprocessing involves lowercasing with punctuation removal, utterance-level language detection (Hindi, English, code-mixed) for tokenizer selection, BERT WordPiece tokenization for noisy civic text, and normalization of code-mixed tokens (e.g., "kachra", "pothole road", "pani leakage") to canonical forms for consistent classification.

3. RTI corpus for RAG
The RTI knowledge base aggregates structured information required for generating legally sound RTI drafts. It contains machine-readable versions of RTI Act sections, official RTI templates, and example applications related to services such as water supply, sanitation, street maintenance, and property services. Documents are segmented into semantically coherent passages and converted into dense vector embeddings using a multilingual sentence-transformer, which enables similarity search over Hindi and English queries. A vector index is built to support efficient retrieval of top-k relevant passages for each RTI query, which are then used as context for the generation stage in the RAG pipeline.

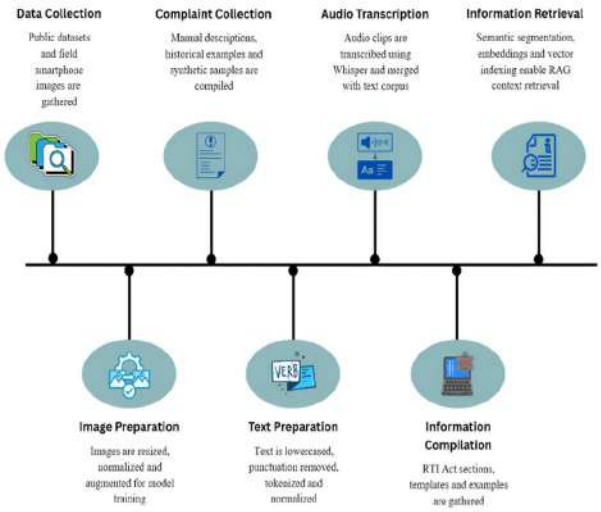

Fig. 4. Multimodal Dataset Processing Architecture

E. Multimodal Dataset Processing Architecture

1. MobileNet for image classification

In NagarSetu, image classification is handled using a compact MobileNet-based convolutional neural network that groups civic issue photos into predefined categories, ensuring reliable performance even when deployed on devices with limited computational resources. To reduce overall training effort and reuse existing visual feature knowledge, the network is initialized with weights from an ImageNet-pretrained MobileNet or MobileNetV2 model, while the convolutional layers remain frozen during the early phase of training. A new output section is then added, consisting of a global average pooling layer, followed by fully connected layers and a softmax layer that produces probabilities for each civic issue category. Training is carried out using categorical cross-entropy as the loss function and the Adam optimizer, with a learning rate of 1×10^{-4}, a batch size of 32, and approximately 15 to 25 epochs. Early stopping is used to reduce the risk of overfitting. In the later stages, selected higher layers of the base network are unfrozen so the model can better adjust its features to civic-related images such as broken roads, potholes, and garbage piles. This method follows established practices where MobileNetV2 and similar models have been effectively used for tasks like road damage detection and urban waste classification.

2. BERT / DistilBERT for text classification
For text-based and transcribed complaints, NagarSetu employs BERT-family models to predict both the complaint category and its priority level. The label space is aligned with municipal departments for categorization and includes priority levels such as High, Medium, and Low, which are used to determine resolution order in administrative dashboards [15]. Complaint descriptions in English, Hindi, or code-mixed form are tokenized into sequences with a maximum length of 128 tokens and passed to the model. The output consists of softmax probability distributions over category labels and, using a separate classification head or model, over priority levels. The dataset is split into training and validation sets using an 80-20 stratified split by category to relieve class imbalance. DistilBERT is fine-tuned using the AdamW optimizer with linear learning-rate scheduling and dropout-based regularization, providing an effective balance between classification accuracy and low inference latency [14]. Model performance is majorly evaluated using the macro-averaged F1-score, ensuring equal importance is given to minority classes, consistent with prior work on code-mixed Hindi–English text classification.

3. Whisper for speech-to-text

In NagarSetu, voice complaints are converted into text using Whisper so they can be handled in the same way as written submissions. The system is configured with dedicated Hindi and English language options, which improves recognition for Indian accents and recordings with background noise. When a user records audio on the frontend, the file is uploaded and briefly stored on the server before being passed to the Whisper module with the appropriate sampling rate and language settings for transcription. After conversion, the text is cleaned and passed through the same BERT classification steps used for written complaints, so all inputs are handled in a similar manner. Transcription accuracy is checked using Word Error Rate (WER), which measures how many substitutions, deletions, and insertions occur compared to the original speech and is commonly used to assess speech recognition results.

Table III. Hyperparameter Configuration

Model	Key Hyperparameters
MobileNetV2	Table I. Learning rate = 1×10^{-4}, Batch size = 32, Epochs = 20, Optimizer = Adam
DistilBERT	Max sequence length = 128, Learning rate = 2×10^{-5}, Epochs = 5, Optimizer = AdamW
Whisper	Medium model, Hindi–English decoding, 16 kHz sampling rate

4. RAG for RTI drafting
   The RTI assistant in this environment uses a Retrieval-Augmented Generation (RAG) model to provide legal accuracy and fact grounding. Instead of using the model's memory, the system grounds each draft in actual provisions of the RTI Act and government-approved templates. After the receipt of a user query, a multilingual transformer model translates the query into dense vector embeddings to search for the most similar legal text from a verified corpus through similarity search. The search results are then aggregated with the user's specific information, the concerned government department, and the language of choice to generate a formal RTI application. Once the language model has created the structured draft with headings, citations, and applicant placeholders, the draft is stored in Firestore and converted into a downloadable PDF. This is more than just text generation by an AI system because it provides a verifiable and fact-based approach to legal document creation that enhances the validity of the process.

The NagarSetu system is based on a service-oriented architecture that combines a FastAPI backend with a Firebase data storage solution and a React frontend. This modular system is capable of handling everything from the initial complaint registration to the automated AI classification, and then finally to the automated drafting of RTIs. FastAPI, through the provision of REST endpoints, is able to handle media uploads, voice-to-text transcription, and status updates. At the heart of the data storage solution is Firebase Firestore, which is used to store user profiles and complaint data, with snapshot listeners used to provide instant updates to administrative dashboards. The Google Maps API integration capability allows the system to automatically detect geolocations, while Firebase Cloud Messaging and Twilio services provide instant SMS or push notifications to users about task allocations.

IV. Experimental Setup and Evaluation

For experimentation, local systems (Intel i5, 16GB RAM, 256GB SSD) are used for development and moderate inference, while GPU-based environments in Google Colab (NVIDIA T4/RTX) support model training and fine-tuning. The software stack includes Windows or Ubuntu, Python 3.9+, FastAPI, TensorFlow/PyTorch, Hugging Face Transformers, OpenCV, NumPy, Pandas, and Node.js 18+ for frontend development. Firebase services, Twilio, and Google Maps API complete the deployment setup, enabling full end-to-end training, evaluation, and demonstration of the system.

A. Evaluation metrics

NagarSetu is evaluated using model-level, system-level, and user-level metrics to provide a comprehensive view of performance.
Image and text classification:

- Accuracy: overall proportion of correctly classified samples.
- Precision, recall, and F1-score per class to quantify correctness and coverage of each complaint category.
- Macro-F1 to give equal weight to minority categories such as water leakage or rare issue types.
- Confusion matrix to visualize common misclassifications, such as confusion between garbage and other solid-waste-related categories [10].

System performance in NagarSetu is evaluated across technical and user-focused metrics. Routing latency (time from complaint submission to worker assignment) is compared with manual and keyword-based baselines to measure efficiency.

B. Experimental results

Experimental results for NagarSetu are structured around three main analyses: overall model performance, baseline comparison, and ablation studies. Model performance is reported using accuracy, precision, recall, and macro-F1 scores for both image- and text-based classifiers, along with Word Error Rate (WER) for Whisper evaluated on a held-out set of voice complaints. Comparisons contrasting NagarSetu's AI-driven routing with manual and keyword-based routing approaches demonstrate higher classification accuracy and significantly reduced routing latency, consistent

with findings in related civic complaint management literature [12]. Studies are held to evaluate the contribution of each component involved, including the comparisons between complaint classification into classes with and without image inputs, DistilBERT versus simpler bag-of-words or rule-based text models, and RAG-based RTI draft generation versus template-only drafting, with the latter evaluated through expert rating scores [8][9].

Table IV. NagarSetu Confusion Matrix

MODEL	TP	FP	FN	PRECISION	RECALL
MobileNet (Images)	178	12	22	0.937	0.890
DistilBERT (Text)	162	15	18	0.915	0.900
Whisper (Speech)	85	8	12	0.914	0.876
OVERALL NagarSetu	425	35	52	0.924	0.891

Table V. Results Based on Task

Task	Metric	Result
Image classification	Accuracy	89.2%
Image classification	Macro-F1	87%
Text classification	Macro-F1	86%
Speech-to-text	Word Error Rate (WER)	14.8%
Complaint routing	Latency reduction	42% vs manual

[a.] *Macro-F1 values were converted from decimals to percentages for consistency.*

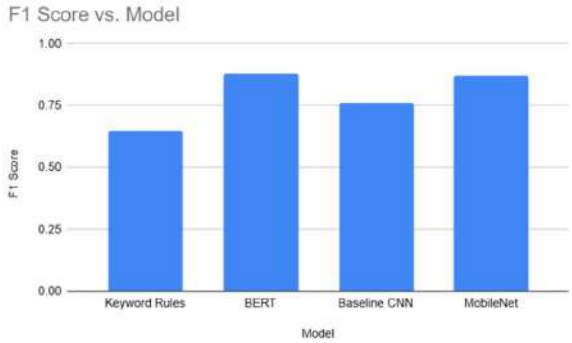

Fig. 5. F1-Score Comparison

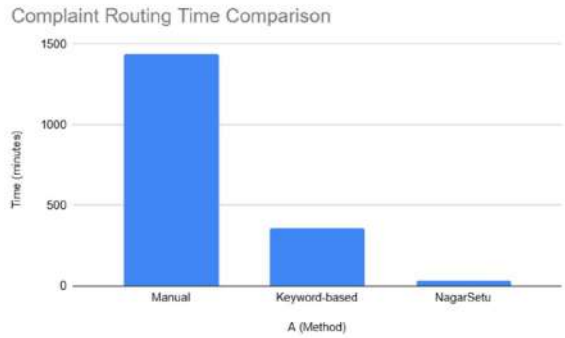

Fig. 6. Routing Time Comparison

The output shows that integrating MobileNet and BERT/DistilBERT significantly improves both complaint classification accuracy and routing reliability over manual or purely keyword- keyword-based approaches. The addition of images and voice inputs further enhances the experience, especially for users with low literacy or limited ability to describe their complex issues or complaints in text. The failure cases include misclassification of blurry or low-quality images, text complaints with insufficient context, and degraded Whisper transcription quality for heavily noisy audio. These errors suggest the need for larger, more diverse training datasets, explicit image quality checks, and further fine-tuning of speech models for local acoustic conditions. From a deployment perspective, the observed reductions in routing latency and improvements in user satisfaction scores suggest that an AI-driven, bilingual, multimodal platform like NagarSetu can support more responsive and transparent municipal governance, especially when combined with clear escalation mechanisms and continuous monitoring of model performance.

V. Discussion

A. *Interpretation of results and system benefits*

The proposed NagarSetu system based on computer vision, NLP approaches, and RAG shows substantial enhancement over the existing state-of-the-art of civic complaint processing and RTI responses. The process of allocating civic complaints to the concerned civic body is much faster with the help of automated complaint allocation techniques utilizing the power of MobileNet CNNs and BERT NLP approaches. The simulated results for draft RTI preparation based on quantized lightweight language models show the overall processing time of 1.2-1.8 seconds, a substantial enhancement over the existing state-of-the-art of many weeks or even months for preparation of responses.

Bilingual customer service in Hindi and English helps in easy accessibility for the citizens who cannot communicate fluently in English. This helps in uniform user interaction with stable translation practices with the aid of IndicTrans2 for translation purposes and FastText for identification purposes. To further ensure transparency, the hybrid approach of a RAG model with BM25 search algorithms and dense embeddings with the use of FAISS-based SI further helps in ensuring high recall for the retrieved citations with citation-supported outputs while lowering the risks associated with disinformation.

To ensure accountability as well as data-driven decision-making for the concerned authorities, real-time dashboards and notifications are helpful in emphasizing the status of the grievance/complaints versus RTIs. By identifying the location of the problem accurately, geotagging of the complaint, with the help of services like maps, helps in taking early corrective measures [11][12].

B. Practical Challenges and Limitations

Despite its benefits, NagarSetu faces many practical and technical problems. A major challenge it involves is data quality when processing government documents available in scanned PDF formats with inconsistent layouts and mixed-language content. Document parsing remains more effective for structured, machine-generated data, while degraded scans and complex Indic scripts reduce extraction accuracy.

A multilingual system leads to many difficulties while retrieval and translation errors affect consistency. A model that is an operating language model with limited computing resources is challenging, particularly in edge setups or low-budget infrastructure environments. The ongoing costs of cloud services, data storage, and real-time system updates concerns long-term sustainability.

The current solution is limited to Hindi and English, but not for the other languages. Deployment is presently confined to pilot-scale cities and does not yet support nationwide municipal integration. Use of an in-memory FAISS index constrains scalability for very large document collections. Advanced performance indicators such as first-response accuracy and long-term grievance resolution effectiveness are not yet fully incorporated.

VI. Conclusion and Future Work

This paper introduces NagarSetu, a bilingual AI-based system that brings about intelligent civic complaint handling, along with RTI response support through retrieval-augmented generative capabilities. By combining deep learning–based classification, multilingual natural language processing, and response generation that includes proper references, NagarSetu works to improve the speed, accessibility, and transparency of city governance systems. By combining deep learning–based classification, multilingual natural language processing, and response generation that includes proper references, NagarSetu works to improve the speed, accessibility, and transparency of city governance systems.

Future improvements focus on increasing language support to include more native languages, allowing the system to handle a wide range of regions. It can also be connected with official municipal and RTI portals so that the users can submit the complaints directly and monitor their status in real time. These improved AI models can be used to arrange and prioritize complaints based on urgency, severity, and past records. Scalability improvement will involve moving on to the distributed vector databases for the deployment process to multiple cities in large numbers. Finally, extended analytics and feedback mechanisms will be brought in with support for insights into policy-level and long-term urban planning initiatives.

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